basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

ELECTRICAL TECHNOLOGY: POWER SYSTEMS

MAY/JUNE 2026

MARKS: 200

TIME: 3 hours

This question paper consists of 19 pages and a 2-page formula sheet.

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SEVEN questions.
2. Answer ALL the questions.
3. Sketches and diagrams must be large, neat and FULLY LABELLED.
4. Show ALL calculations and round off answers correctly to TWO decimal places.
5. Number the answers correctly according to the numbering system used in this question paper.
6. You may use a non-programmable calculator.
7. Calculations MUST include:
   - 7.1 Formulae and manipulations where needed
   - 7.2 Correct replacement of values
   - 7.3 Correct answer and relevant units where applicable
8. A formula sheet is attached at the end of this question paper.
9. Write neatly and legibly.

QUESTION 1: MULTIPLE-CHOICE QUESTIONS

Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question numbers (1.1 to 1.15) in the ANSWER BOOK, e.g. 1.16 D.

1.1 An employee with a well-developed sense of teamwork …

A works well independently without any distractions.
B encourages open communication and collaboration with others.
C enjoys having a strict leader who makes all the decisions. (1)

1.2 The inductive reactance of an inductor … with an increase in frequency.

A increases linearly
B decreases linearly
C increases exponentially
D decreases exponentially (1)

1.3 The ratio of the resistance to the impedance in an RLC series circuit is known as the …

A apparent power.
B Q-factor.
C power factor.
D real power. (1)

1.4 The bandwidth of a resonant circuit will … with an increase in the Q-factor.

A decrease
B increase
C remain unchanged
D sum to zero (1)

1.5 The … is not consumed, but oscillates between the supply and the load component.

A active power
B apparent power
C reactive power
D power loss (1)

1.6 Transformers at power stations … in transmission lines.

A step down the voltage to reduce iron losses
B step down the voltage to reduce I^2R losses
C step up the voltage to reduce iron losses
D step up the voltage to reduce I^2R losses (1)

1.7 To correctly measure the total power when using two wattmeters in a three-phase system, do the following:

(i) Connect the current coil for each meter in series with a phase.
(ii) Connect a terminal of each potential coil to the same phase as the current coil.
(iii) Connect the other terminal of both potential coils to the third phase.

A (i) and (ii) only
B (ii) and (iii) only
C (i) only
D All the above-mentioned (1)

1.8 … is an external condition which could cause faults to develop in three-phase transformers.

A Overloading
B Switching surges
C Lightning
D All the above-mentioned (1)

1.9 The highest contributor of heat losses in a transformer is …

A hysteresis losses.
B eddy current losses.
C copper losses.
D stray losses. (1)

1.10 The difference between the rotating stator field and the rotor speed is known as …

A efficiency.
B slip.
C synchronous speed.
D commutation. (1)

1.11 A continuity test on the coils of an induction motor is done by connecting …

A a multimeter setting on the Ω-scale at the ends of each coil.
B a multimeter setting on the MΩ-scale at the ends of each coil.
C an insulation resistance tester setting on the Ω-scale at the ends of two different coils.
D an insulation resistance tester setting on the MΩ-scale between the ends of each coil and earth. (1)

1.12 The process by which the contactor coil remains energised after the start button is released, is called …

A locking out.
B simulating.
C timing.
D latching. (1)

1.13 An electromechanical device that converts a mechanical force into an electrical signal is known as a …

A level sensor.
B strain gauge.
C temperature sensor.
D proximity sensor. (1)

1.14 The … executes the programmed instructions during the PLC scan cycle.

A input scan
B process scan
C output scan
D All the above-mentioned (1)

1.15 A/An … is used to dissipate the excessive energy in the form of heat during the process of regenerative braking.

A braking resistor
B DC capacitor
C IGBT transistor
D free-wheeling diode (1)
[15]

QUESTION 2: OCCUPATIONAL HEALTH AND SAFETY

2.1 State whether it is an *unsafe act* or a *condition* when a person neglects to wear safety goggles while working with a pedestal drill. (1)

2.2 Define the term *employer* with reference to the Occupational Health and Safety Act, 1993 (Act 85 of 1993). (2)

2.3 Explain *quantitative risk analysis*. (2)

2.4 Explain the purpose of human rights in the workplace. (2)

2.5 An employee who has a sense of responsibility displays good work ethics. State how it can help the company achieve the production goals. (3)
[10]

QUESTION 3: RLC CIRCUITS

3.1 Explain the following with reference to RLC circuits:

3.1.1 Impedance (2)

3.1.2 Resonant frequency (2)

3.1.3 The status of current and impedance at resonance in a parallel circuit (2)

3.2 FIGURE 3.2 below shows an RLC circuit series connected to an AC supply with voltmeter readings across the inductor and capacitor.

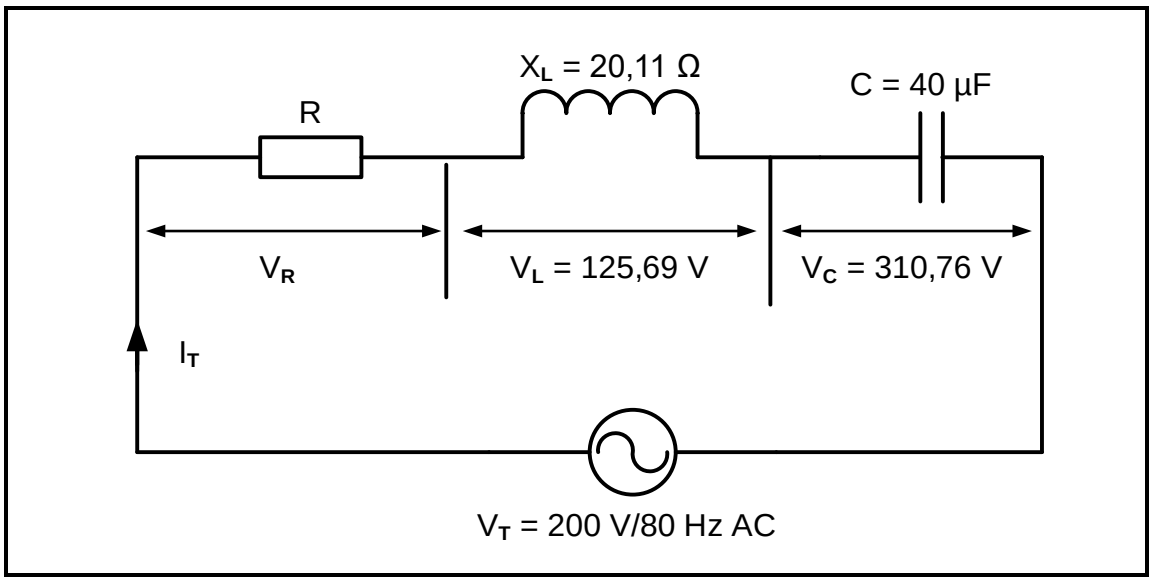


FIGURE 3.2: RLC SERIES CIRCUIT

Given:

X_L = 20,11 Ω
Z = 31,98 Ω
C = 40 µF
V_T = 200 V
V_L = 125,69 V
V_C = 310,76 V
f = 80 Hz

3.2.1 State whether the current through the inductor leads or lags the voltage across it. (1)

Calculate the following:

3.2.2 The capacitive reactance (3)

3.2.3 The value of the inductor (3)

3.2.4 The value of the resistor if the impedance is 31,98 ohms (3)

3.2.5 The total current (I_T) through the circuit (3)

3.2.6 The resonant frequency (3)

3.3 Refer to QUESTION 3.2 on the previous page to redraw and complete the voltage phasor diagram in FIGURE 3.3 below in your ANSWER BOOK.

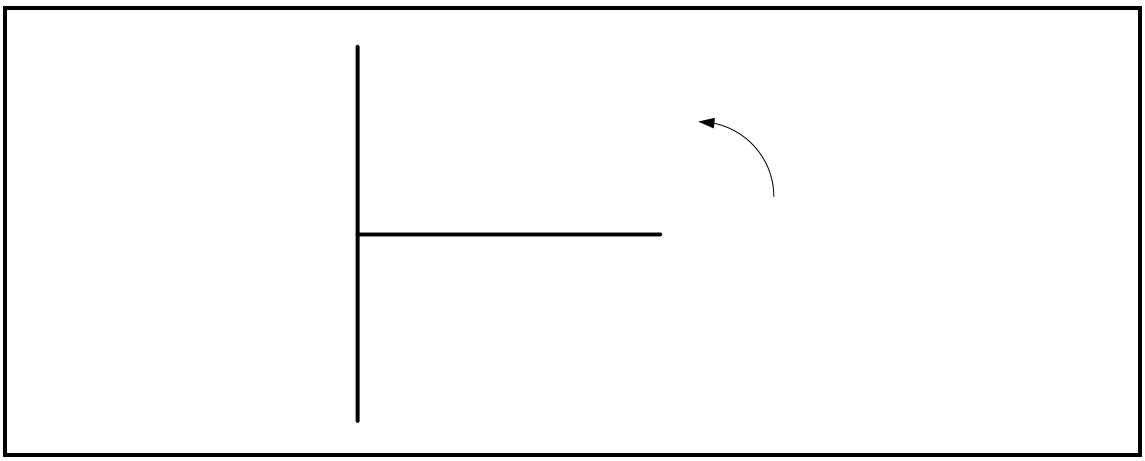


FIGURE 3.3: INCOMPLETE PHASOR DIAGRAM (4)

3.4 FIGURE 3.4 below shows a phasor diagram with the current flowing through the components of an RLC parallel circuit connected to an AC supply. Answer the questions that follow.

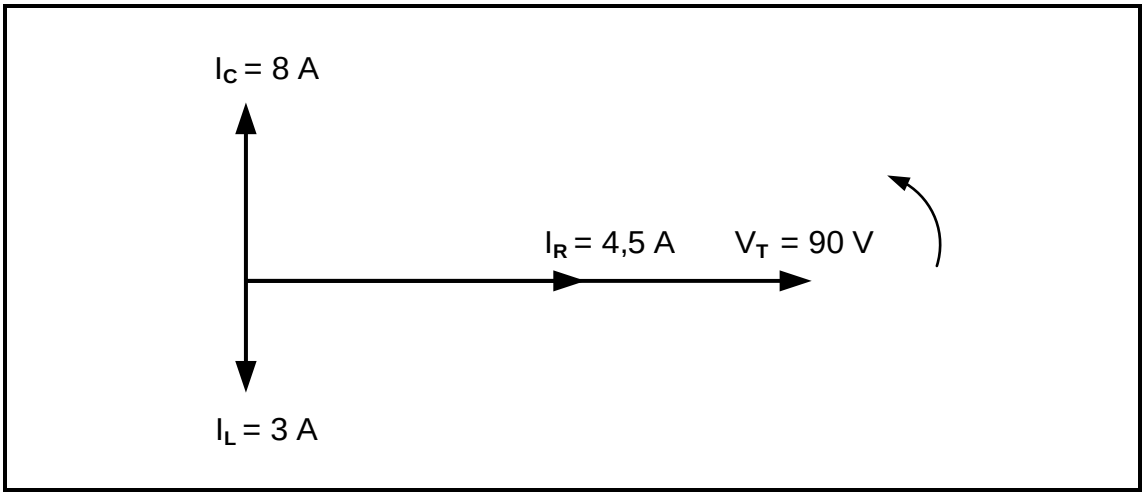


FIGURE 3.4: RLC PARALLEL CIRCUIT PHASOR DIAGRAM

Given:

I_C = 8 A
I_L = 3 A
I_R = 4,5 A
V_T = 90 V
X_C = 11,25 Ω
fr = 150 Hz

Calculate the following:

3.4.1 Inductive reactance (3)

3.4.2 The value of the capacitor if the capacitive reactance is 11,25 Ω (3)

3.4.3 The total current in the circuit (3)

[35]

QUESTION 4: THREE-PHASE AC GENERATION

4.1 Name TWO environmentally friendly methods of power generation in South Africa that reduces air pollution. (2)

4.2 Name ONE advantage of three-phase distribution over single-phase distribution of power. (1)

4.3 State TWO advantages of power factor correction for the consumer. (2)

4.4 Refer to FIGURE 4.4 below and explain how three-phase supply is generated.

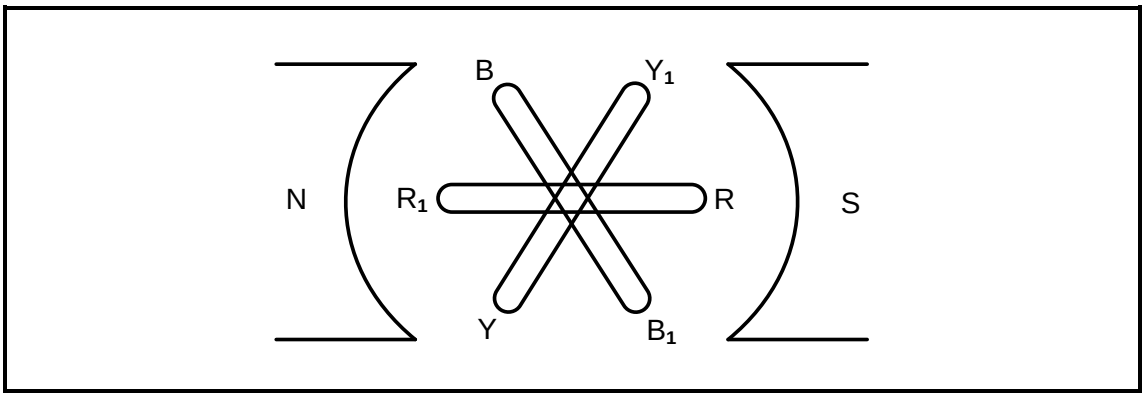


FIGURE 4.4: THREE-PHASE GENERATOR (3)

4.5 FIGURE 4.5 below shows a 20 kVA star-connected alternator that powers a load with a power factor of 0,85 lagging. The phase voltage is 230 V. Answer the questions that follow.

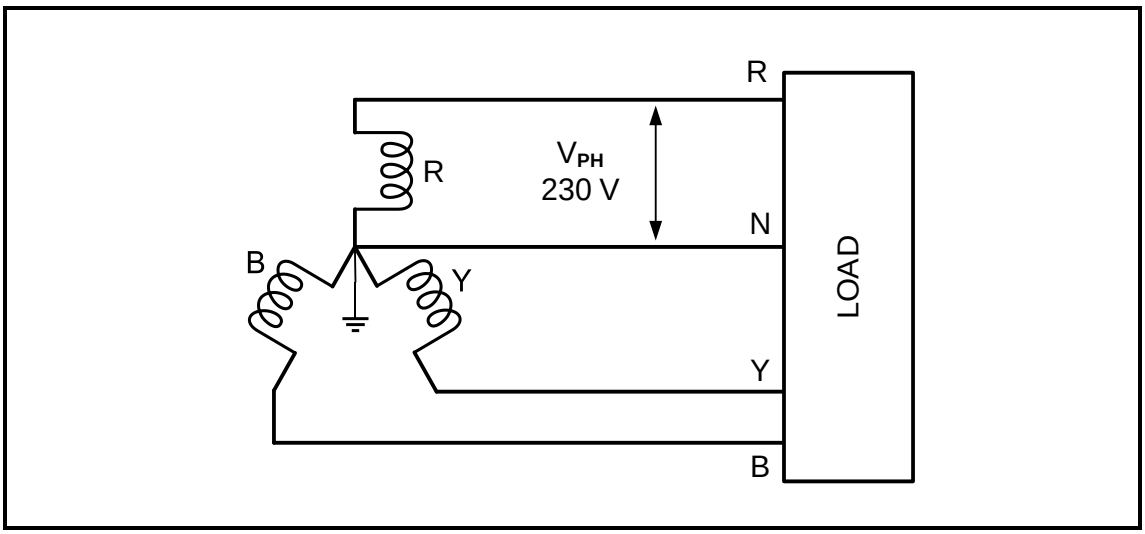


FIGURE 4.5: THREE-PHASE AC GENERATOR

Given:

S = 20 kVA
V_{PH} = 230 V
pf = 0,85

Calculate the following:

4.5.1 Line voltage (3)

4.5.2 Line current (3)

4.5.3 Active power (3)

4.5.4 Reactive power at a phase angle of 31,79° (3)

4.6 FIGURE 4.6 below shows three wattmeters connected to a load. Answer the questions that follow.

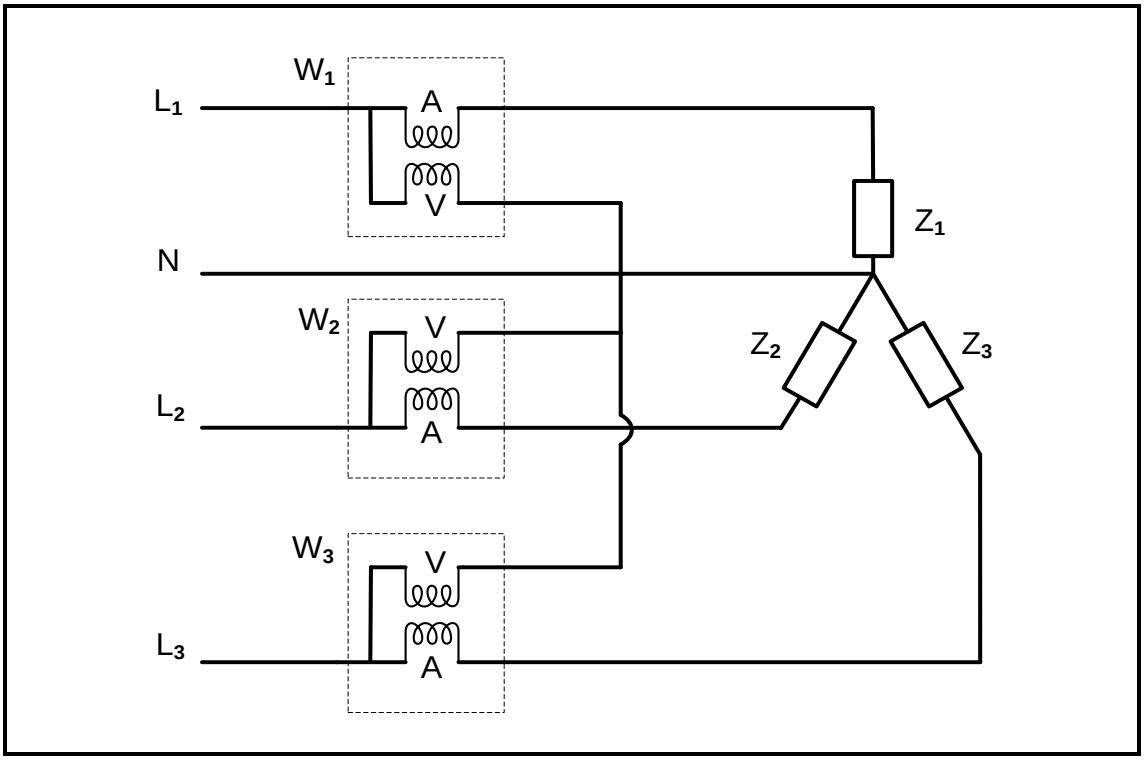


FIGURE 4.6: THREE-WATTMETER METHOD

4.6.1 Explain why a voltmeter and an ammeter CANNOT be used to measure power in FIGURE 4.6 above. (1)

4.6.2 Calculate the power measured by W_2 if W_1 measured 2 010 W, W_3 measured 670 W and the total power measured is 2 747 W. (3)

4.6.3 State the difference between a *wattmeter* and a *kilowatt-hour meter*. (2)

4.6.4 Determine the value of the current in the neutral conductor when loads Z_1, Z_2 and Z_3 have equal impedance. (1)

4.7 Describe the effects of a low power factor in the generation and supply of a three-phase AC supply. (4)

4.8 FIGURE 4.8 below shows the connection of an instrument connected to a three-phase symmetrically loaded network. A reading of 0,9 lagging is displayed on its screen. Answer the questions that follow.

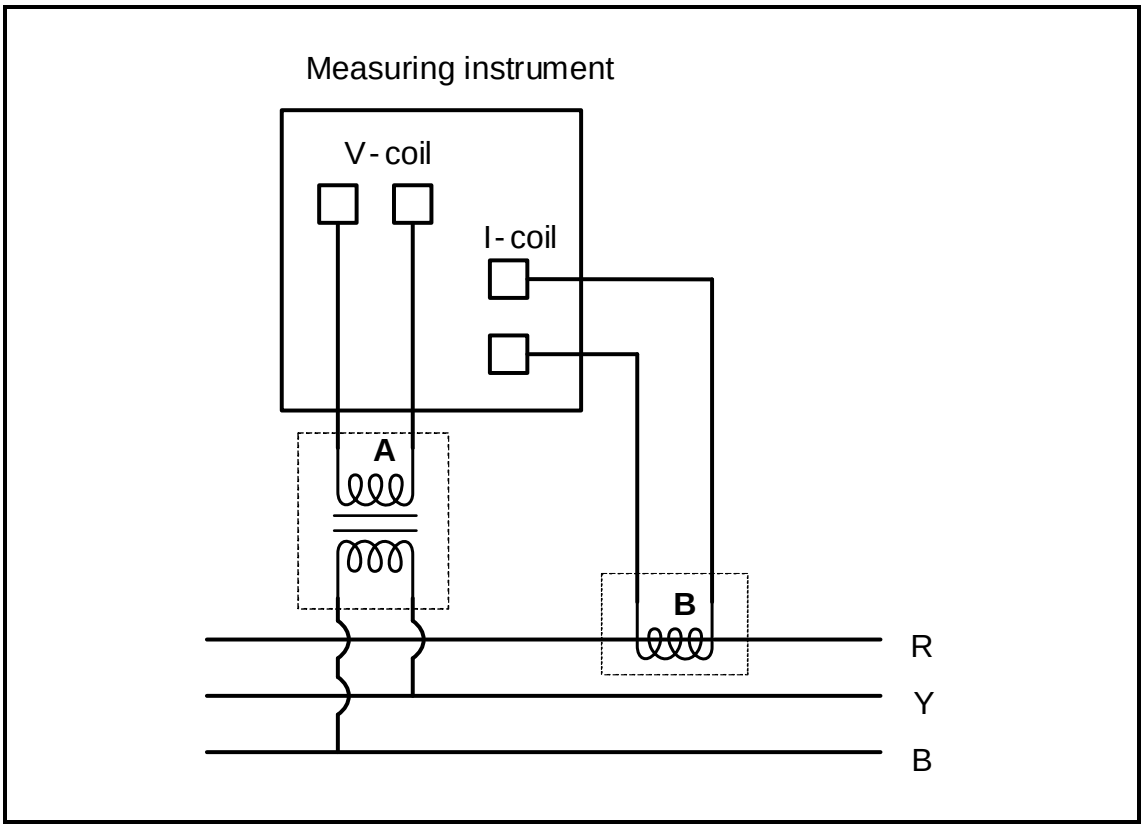


FIGURE 4.8: MEASURING DEVICE

4.8.1 Identify the measuring instrument in FIGURE 4.8 above. (1)

4.8.2 Name devices **A** and **B** in FIGURE 4.8 above. (2)

4.8.3 State whether the addition of inductive loads will increase or decrease the reading displayed on the instrument. (1)
[35]

QUESTION 5: THREE-PHASE TRANSFORMERS

5.1 Describe the term *transformation* with reference to transformers. (2)

5.2 Refer to FIGURE 5.2 below and answer the questions that follow

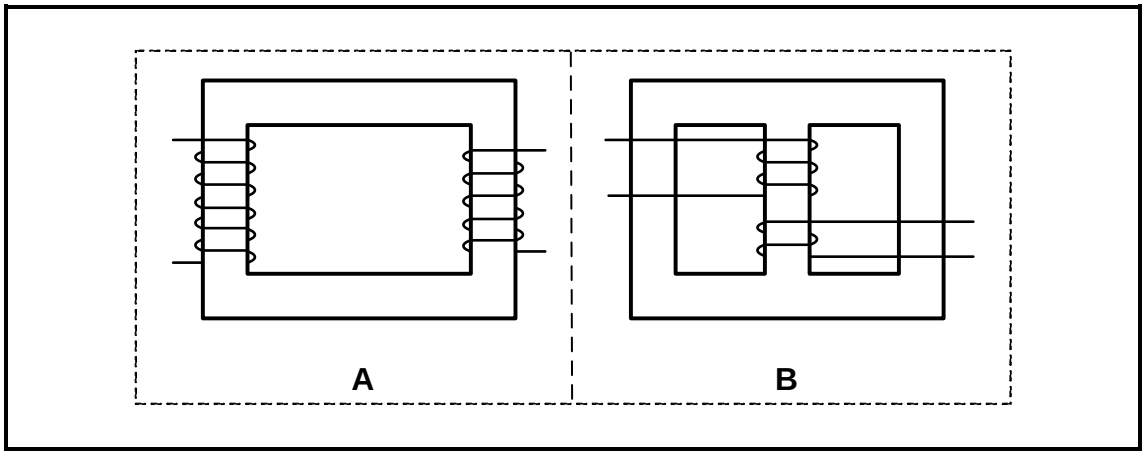


FIGURE 5.2: THREE-PHASE TRANSFORMER

5.2.1 Identify the type of transformer construction used in **A** and **B** in FIGURE 5.2 above. (2)

5.2.2 Explain why an iron core is needed in the construction of a transformer. (2)

5.3 Name TWO cooling methods used in oil-immersed transformers. (2)

5.4 State TWO properties that must be identical in single-phase transformers for them to be used as a three-phase transformer unit. (2)

5.5 A 300 kW delta-connected load with a power factor of 0,86 is connected to a delta-star transformer. The primary and the secondary line voltages of the transformer are 5 kV and 480 V respectively.

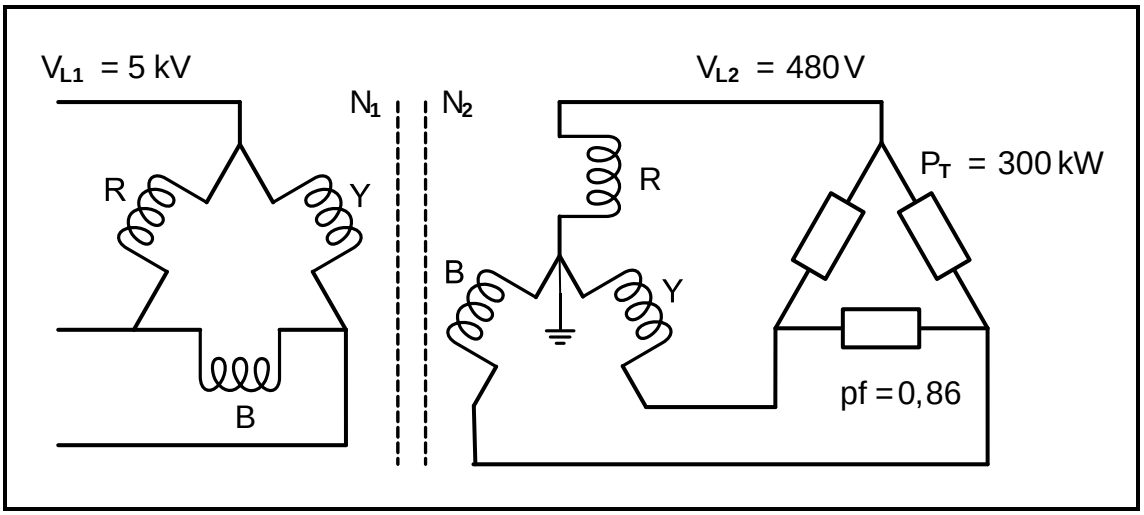


FIGURE 5.5: DELTA-STAR TRANSFORMER

Given:

P_{IN} = 300 kW
V_{L1} = 5 kV
V_{L2} = 480 V
pf = 0,86
η = 85%

Calculate the following:

5.5.1 Secondary line current (3)

5.5.2 Phase current of the load (3)

5.5.3 Apparent power (3)

5.5.4 Primary line current (3)

5.5.5 Turns ratio of the transformer if V_{PH2} = 277,13 V (3)

5.6 Refer to FIGURE 5.6 below and answer the questions that follow.

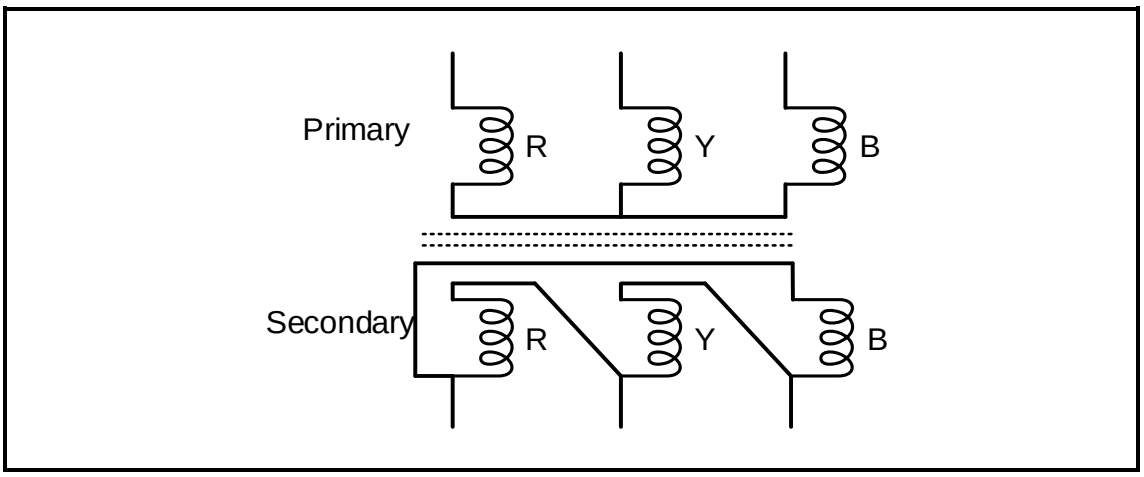


FIGURE 5.6: SINGLE-PHASE TRANSFORMERS CONNECTED AS A THREE-PHASE UNIT

5.6.1 State ONE disadvantage of connecting single-phase transformers as a three-phase unit as shown in FIGURE 5.6 above. (1)

5.6.2 State ONE application of the transformer unit in FIGURE 5.6 above. (1)

5.6.3 Explain why the windings in a three-phase transformer constructed out of three single-phase transformers must be connected in the correct sequence. (3)

[30]

QUESTION 6: THREE-PHASE MOTORS AND STARTERS

6.1 Explain the concept *induction* with reference to three-phase induction motors. (3)

6.2 Refer to the speed of induction motors and answer the questions that follow.

6.2.1 Explain *full-load speed*. (2)

6.2.2 Calculate the synchronous speed of a three-phase motor with four pole pairs per phase that are connected to a 400 V/50 Hz supply. (3)

6.2.3 Calculate the percentage slip if the rotor turns at 725 r/min. (3)

6.3 Name TWO mechanical inspections that could be carried out on the rotor and bearings of a motor. (2)

6.4 A three-phase delta-connected motor draws a line current of 5 A from a 380 V/50 Hz supply and the phase angle is 20°. The efficiency of the motor is 90%.

Given:

I_L = 5 A
V_L = 380 V
θ = 20°
η = 90%
f = 50 Hz

Calculate the following:

6.4.1 Power factor (3)

6.4.2 Input power (3)

6.4.3 Total losses (3)

6.5 FIGURE 6.5 below shows a three-phase motor terminal box connected in a specific configuration. Redraw the terminal box in the ANSWER BOOK and show how the direction of motor rotation can be reversed.

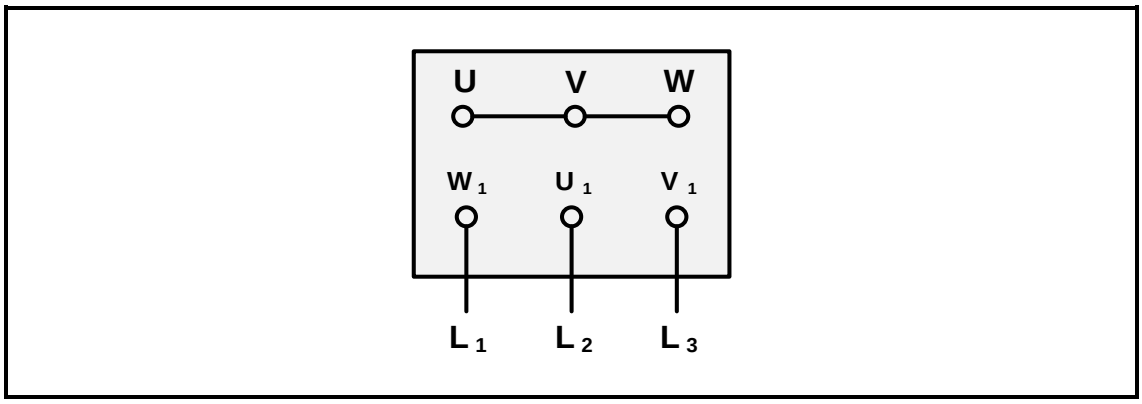


FIGURE 6.5: THREE-PHASE MOTOR TERMINAL BOX (1)

6.6 FIGURE 6.6 below shows the control circuit of an automatic star-delta starter. The timer is set to 5 seconds. Answer the questions that follow.

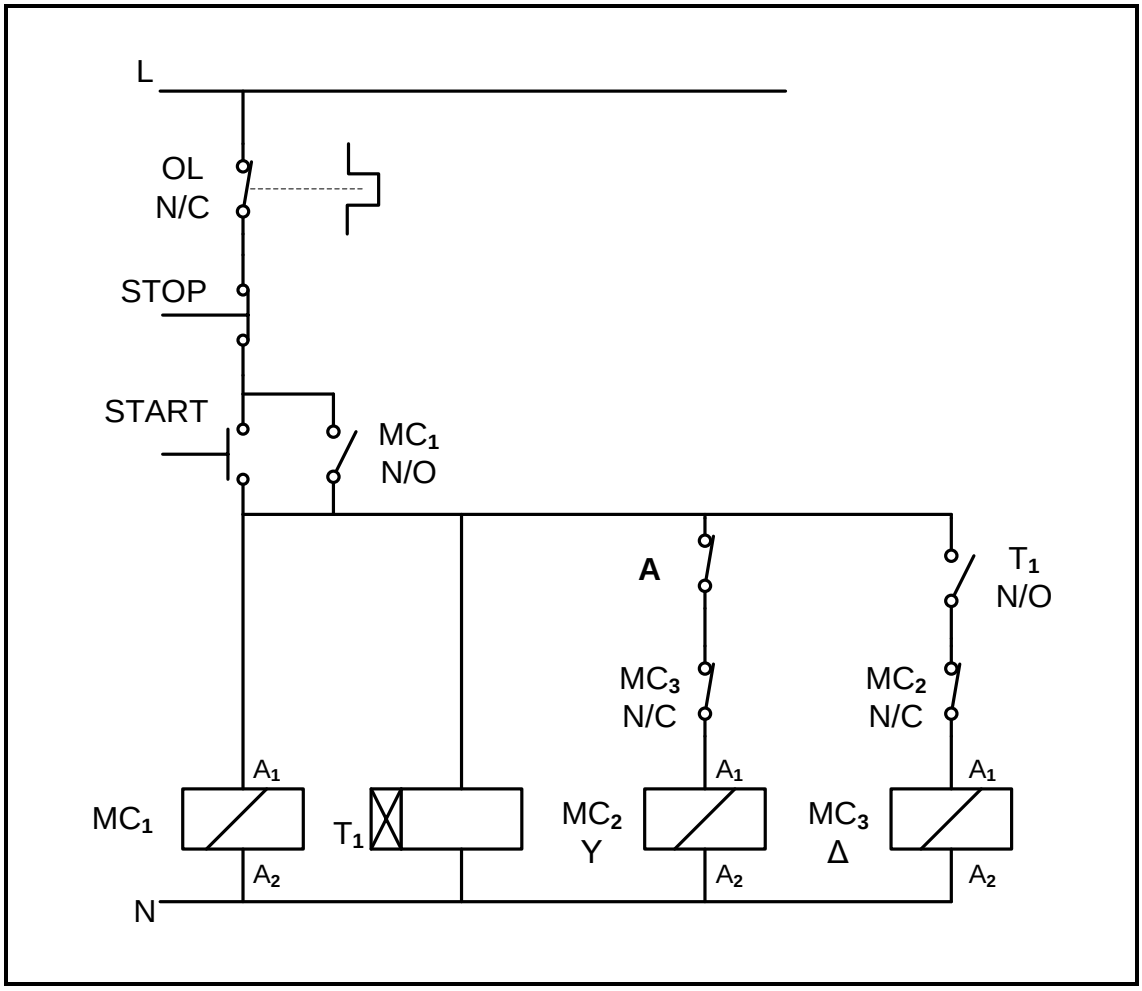


FIGURE 6.6: AUTOMATIC STAR-DELTA CONTROL CIRCUIT

6.6.1 Identify contact **A**. (1)

6.6.2 Explain the function of MC_3N/C in FIGURE 6.6 above. (2)

6.6.3 State TWO functions of the automatic star-delta control circuit in FIGURE 6.6 above. (2)

6.6.4 Describe the operation of the circuit from when the start button is pressed. (7)
[35]

QUESTION 7: PROGRAMMABLE LOGIC CONTROLLERS (PLCs)

7.1 State TWO disadvantages of a relay control system in comparison with PLC control systems. (2)

7.2 State TWO applications of a proximity sensor. (2)

7.3 Explain why a soft-wired system is cheaper to implement than a hardwired system. (2)

7.4 Name ONE device other than a relay that is used in the output module of a PLC. (1)

7.5 Differentiate between a *hard-wired contactor* and a *soft-wired contactor* used within the software program of a PLC. (4)

7.6 Refer to FIGURE 7.6 below and answer the questions that follow.

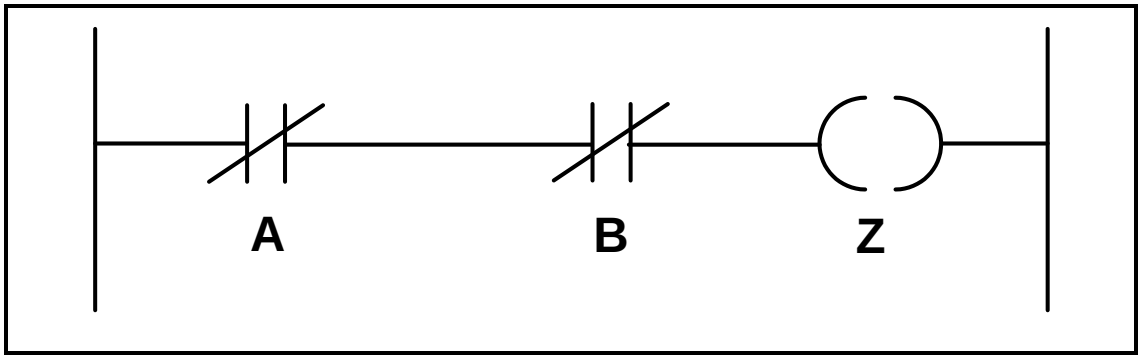


FIGURE 7.6: LADDER LOGIC DIAGRAM

7.6.1 Identify the logic gate represented by the ladder logic diagram in FIGURE 7.6 above. (1)

7.6.2 Draw the symbol of the logic gate represented by the ladder logic diagram in FIGURE 7.6 above. (2)

7.6.3 Complete the truth table of the logic gate represented by FIGURE 7.6 above by writing down only the state of the output in the truth table below.

A	B	OUTPUT
1	0	(X)
1	1	(Y)

FIGURE 7.6.3: TRUTH TABLE (2)

7.7 Answer the following questions with reference to timers in PLC programming.

7.7.1 Explain how an OFF-delay timer creates the time delay when de-energised. (3)

7.7.2 Redraw the PLC ladder logic diagram in FIGURE 7.7.2 below in the ANSWER BOOK and complete it to perform an ON-delay function. Output Q is activated by a switch and output Q must only turn ON 3 seconds after activating the switch.

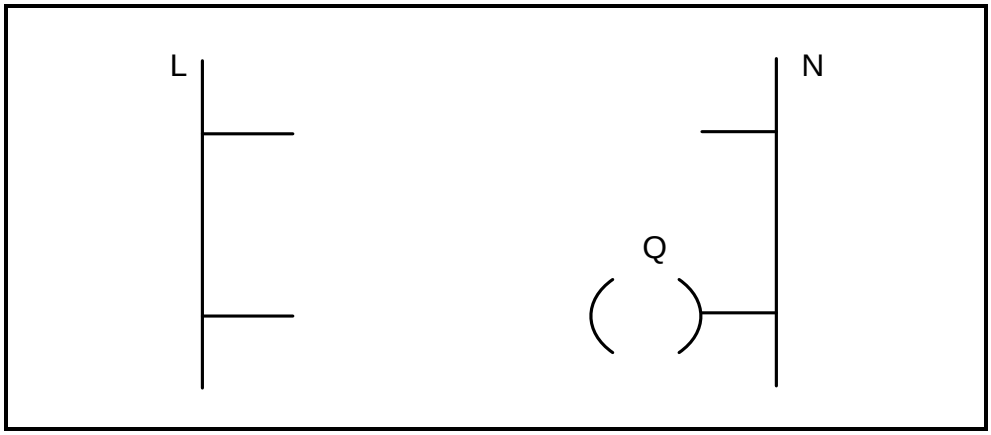


FIGURE 7.7.2: INCOMPLETE PLC LADDER LOGIC DIAGRAM (3)

7.8 FIGURE 7.8(A) below shows the circuit diagram of a DOL starter with indicating lamps.

Redraw and complete the PLC ladder logic diagram in FIGURE 7.8(B), that will execute the same function in a PLC system, in the ANSWER BOOK.

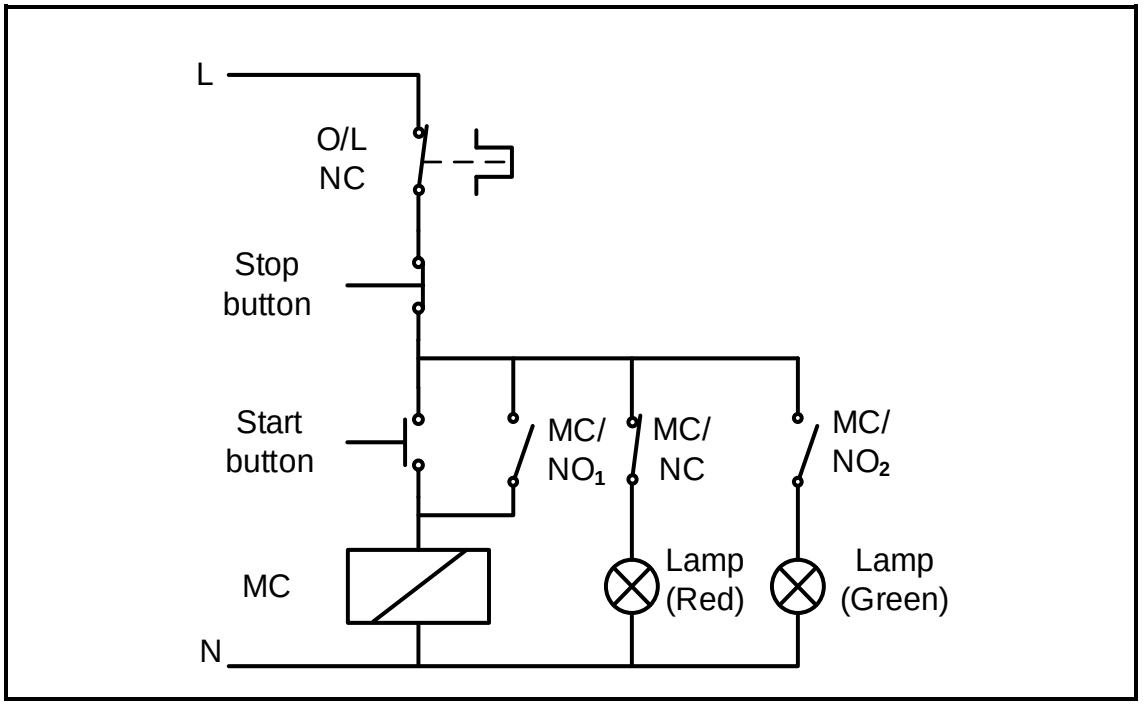


FIGURE 7.8(A): DOL STARTER WITH INDICATING LAMPS

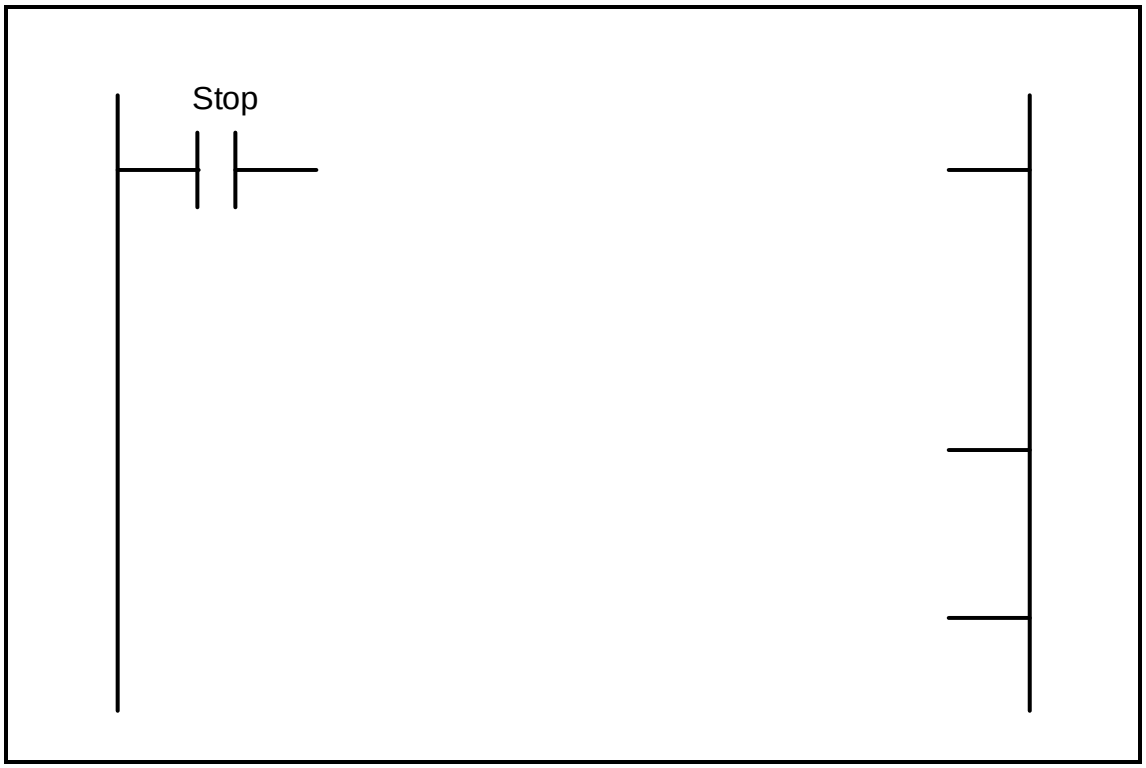


FIGURE 7.8(B): INCOMPLETE PLC LADDER LOGIC DIAGRAM (8)

7.9 Refer to regenerative braking and answer the questions that follow.

7.9.1 Give TWO examples where this braking method is used. (2)

7.9.2 Describe *regenerative energy*. (3)

7.10 Answer the following questions with reference to variable speed drives (VSDs).

7.10.1 Name TWO main parts of the VSD, other than the inverter. (2)

7.10.2 State TWO advantages of using a VSD as a motor controller. (2)

7.10.3 Name ONE type of motor used with a VSD. (1)
[40]

TOTAL: 200

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FORMULA SHEET

RLC CIRCUITS

$P = V\,I\,\text{Cos}\,\theta$

$X_L = 2\pi fL$

$X_C = \frac{1}{2\pi fC}$

$f_r = \frac{1}{2\pi\sqrt{LC}}$ **OR** $f_r = \frac{f_1 + f_2}{2}$

$BW = \frac{f_r}{Q}$ **OR** $BW = f_2 - f_1$

SERIES

$V_R = IR$

$V_L = IX_L$

$V_C = IX_C$

$I_T = \frac{V_T}{Z}$ **OR** $I_T = I_R = I_C = I_L$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$V_T = \sqrt{V_R{}^2 + (V_L - V_C)^2}$ **OR** $V_T = IZ$

$\text{Cos}\,\theta = \frac{R}{Z}$ **OR** $\text{Cos}\,\theta = \frac{V_R}{V_T}$

$Q = \frac{X_L}{R} = \frac{X_C}{R} = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{1}{R}\sqrt{\frac{L}{C}}$

PARALLEL

$V_T = V_R = V_C = V_L$

$I_R = \frac{V_T}{R}$

$I_C = \frac{V_T}{X_C}$

$I_L = \frac{V_T}{X_L}$

$I_T = \sqrt{I_R^2 + (I_L - I_C)^2}$

$Z = \frac{V_T}{I_T}$

$\text{Cos}\,\theta = \frac{I_R}{I_T}$

$Q = \frac{R}{X_L} = \frac{R}{X_C}$

THREE-PHASE AC GENERATION

STAR

$V_L = \sqrt{3}\,V_{PH}$

$V_{PH} = I_{PH}Z_{PH}$

$I_L = I_{PH}$

DELTA

$V_L = V_{PH}$

$V_{PH} = I_{PH}Z_{PH}$

$I_L = \sqrt{3}\,I_{PH}$

POWER

$S(P_{app}) = \sqrt{3}\,V_L\,I_L$

$Q(P_r) = \sqrt{3}\,V_L\,I_L\text{Sin}\,\theta$

$P = \sqrt{3}\,V_L I_L \text{Cos}\,\theta$

$\text{Cos}\,\theta = \frac{P}{S}$

EFFICIENCY

$\eta = \frac{P_{OUT}}{P_{IN}} \times 100$

TWO-WATTMETER METHOD

$P_T = P_1 + P_2$

$\text{Tan}\,\theta = \sqrt{3}\left(\frac{P_1 - P_2}{P_1 + P_2}\right)$

THREE-WATTMETER METHOD

$P_T = P_1 + P_2 + P_3$

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THREE-PHASE TRANSFORMERS

STAR

$V_L = \sqrt{3}\, V_{PH}$

$I_L = I_{PH}$

DELTA

$V_L = V_{PH}$

$I_L = \sqrt{3}\, I_{PH}$

POWER

$S(P_{app}) = \sqrt{3}\, V_L\, I_L$

$Q(P_r) = \sqrt{3}\, V_L\, I_L \text{Sin}\, \theta$

$P = \sqrt{3}\, V_L\, I_L \text{ Cos } \theta$

$\text{Cos } \theta = \frac{P}{S}$

$\frac{V_{PH(1)}}{V_{PH(2)}} = \frac{N_1}{N_2} = \frac{I_{PH(2)}}{I_{PH(1)}}$

$\text{Turns ratio: TR} = \frac{N_1}{N_2}$

EFFICIENCY

$\eta = \frac{P_{OUT}}{P_{OUT} + \text{losses}} \times 100$

THREE-PHASE MOTORS AND STARTERS

STAR

$V_L = \sqrt{3}\, V_{PH}$

$I_L = I_{PH}$

DELTA

$V_L = V_{PH}$

$I_L = \sqrt{3}\, I_{PH}$

POWER

$S(P_{app}) = \sqrt{3}\, V_L\, I_L$

$Q(P_r) = \sqrt{3}\, V_L\, I_L \text{Sin}\, \theta$

$P = \sqrt{3}\, V_L\, I_L \text{ Cos } \theta$

$P = \sqrt{3}\, V_L\, I_L \text{ Cos } \theta\, \eta$

$\text{Cos } \theta = \frac{P}{S}$

EFFICIENCY

$\eta = \frac{P_{IN} - \text{losses}}{P_{IN}} \times 100$

$\eta = \frac{P_{OUT}}{P_{OUT} + \text{losses}} \times 100$

$\eta = \frac{P_{OUT}}{P_{IN}} \times 100$

SPEED

$\text{(pole pairs) } p = \frac{\text{poles per phase}}{2}$

$n_s = \frac{60\, f}{p}$

$\text{Per unit slip} = \frac{n_s - n_r}{n_s}$

$\%\text{ slip} = \frac{n_s - n_r}{n_s} \times 100$

$n_r = n_s(1 - \%\text{ slip})$

$\text{Slip} = n_s - n_r$

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